CSCI250 Computer Organization

Data Representations and Computer Arithmetic

August 18, 2023

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   - Lab Instructions
   - Learning Objectives
2. Binary Numbers
   - A binary card trick
   - Analog to Digital
3. Base 16
   - Hexadecimal Notation
   - Decimal to Hexadecimal

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Instructions for Lab Answers

- Open the provided document for Lec04.txt in Notepad or Notepad++ or the equivalent for a Mac.
- I want a .txt document not anything else!!!
- I will show you how to work a problem and then ask you to work several.
- Answer each question as you get to them in the slides - begin your answer on the line beneath the question and use tabs to indent each line of the answer.

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Learning Objectives

- Understand how computers represent data.
- Convert integer values to binary and back.
- Convert binary values to hexadecimal and back.
- Convert floating point values to binary and back.
- understand ways to represent signed values in binary.
- Perform addition, subtraction, and multiplication in binary.

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Binary Card Trick

- Download and unzip the BinaryCardTrick.zip file.
- Think of a number between 1 and 31
- Mark the cards that have the number you chose on them
- Run the program several times and try to figure out how it works.

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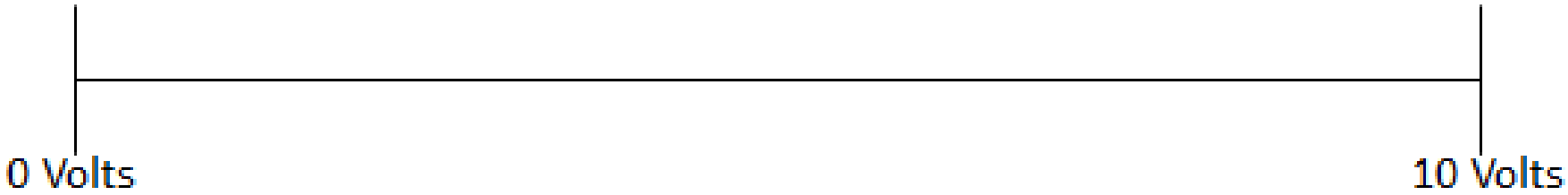
From Digital Signals to Binary 1

- Computers are digital devices: interpret electrical signals as having discrete values.
- Binary has two discrete values 0 and 1
- An electrical signal is an analog quantity and is measured over a continuous (infinite) range of voltage values.



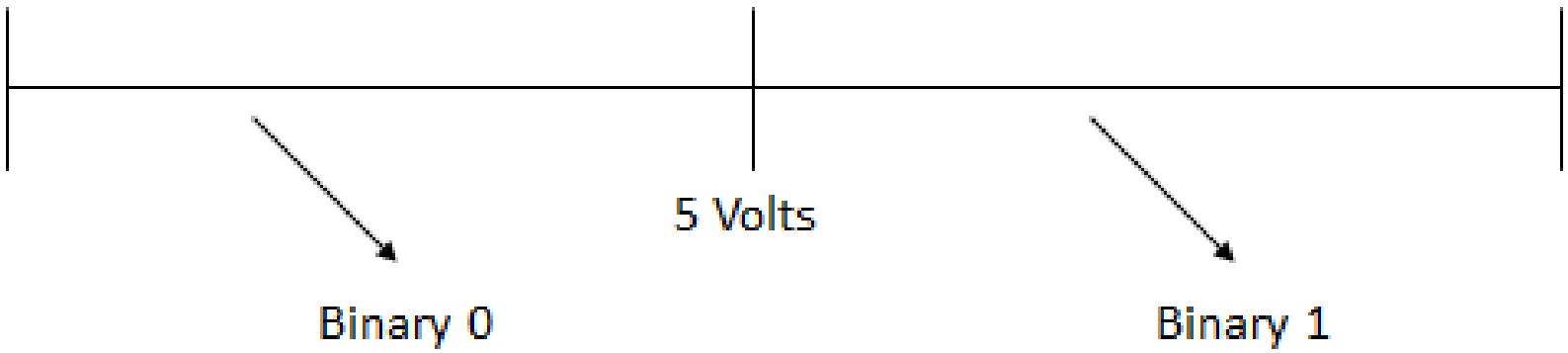
- For example, this electrical signal may have any voltage between 0 and 10 volts.

From Digital Signals to Binary 2

- Why not just use analog?
- Harder to reliably store and process analog values which can assume an infinite number of possible values over a given interval.
- Analog values are subject to noise making difficult to reliably distinguish whether or not two nearby analog values should be considered identical or not.

From Digital Signals to Binary 3

- Convert an analog signal to binary by mapping the continuous range of voltages into 0 or 1.
- High voltages are assigned the binary value 1.
- Low voltages are assigned the binary value 0.
- Example: How should the 0/1 voltage threshold be determined?

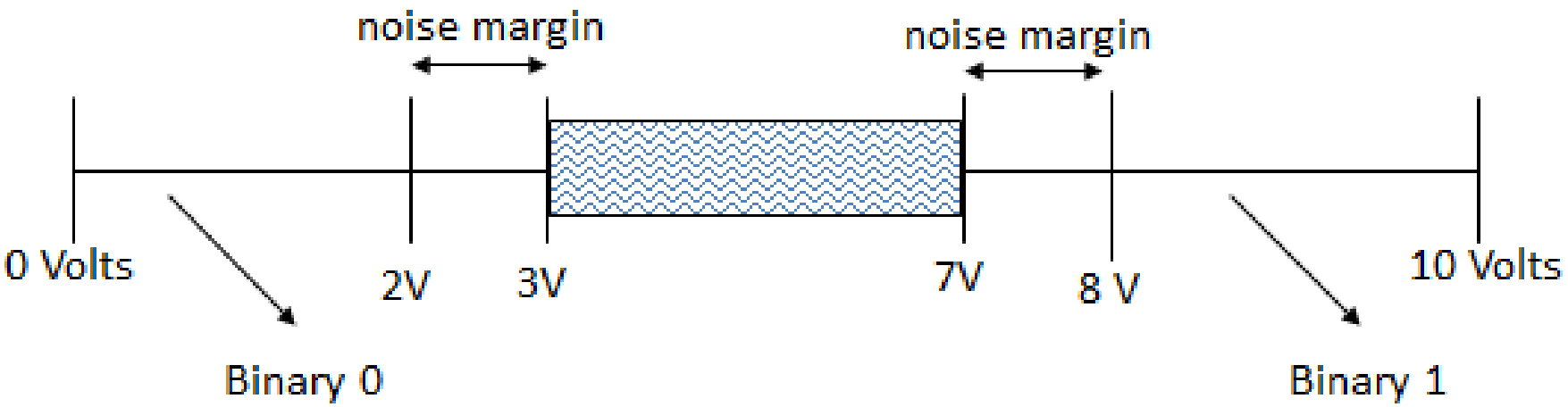


From Digital Signals to Binary 4

- Must use spaced threasholds for reliable analog-to-digital

- Voltages between 0 and 2 are guaranteed to produce 0.
- Voltages between 8 and 10 are guaranteed to produce 1.

From Digital Signals to Binary 5



- Voltages between 2 and 3 will also be interpreted as 0.
- Voltages between 7 and 8 will also be interpreted as 1.
- Voltages between 3 adn 7 fall in the "forbidden zone" and cannot be reliably assigned binary values

Binary Representation of Positive Integers

- Binary number use a *base 2* number system.
- First consider the familiar *base 10* decimal number system.
  $1543 = (1x10^3) + (5x10^2) + (4x10^1) + (3x10^0)$
- A binary number is written as a string of 0 or 1 binary digits (bits)

$$101b = (1x2^2) + (0x2^1) + (1x2^0)$$

Decimal	Binary
0	000 b
1	001 b
2	010 b
3	011 b
4	100 b
5	101 b
6	110 b
7	111 b

We can represent 8 values with 3 bits ranging from 0 to 7.

Binary Representation of Positive Integers

4-bits are called a nibble. 4-bit representations of positive base 10 integers. We can represent 16 numbers from 0 to 15 with 4 bits. This will be an important point when we talk about hexadecimal numbers.

Decimal	Binary
0	0000 b
1	0001 b
2	0010 b
3	0011 b
4	0100 b
5	0101 b
6	0110 b
7	0111 b

Decimal	Binary
8	1000 b
9	1001 b
10	1010 b
11	1011 b
12	1100 b
13	1101 b
14	1110 b
15	1111 b

Range of binary values

Number of bits	Number of Possible Values
1	2
2	4
3	8
4	16

Now let's generalize:

Q1 How many values can we represent using n bits?

Low Order & High Order Bits

- The *low order bit* corresponds to the integer value 2^0 when that bit is set (has a value of 1).
- The *high order bit* corresponds to the integer value 2^{n-1} where n is the number of bits in the binary number when that bit is set.

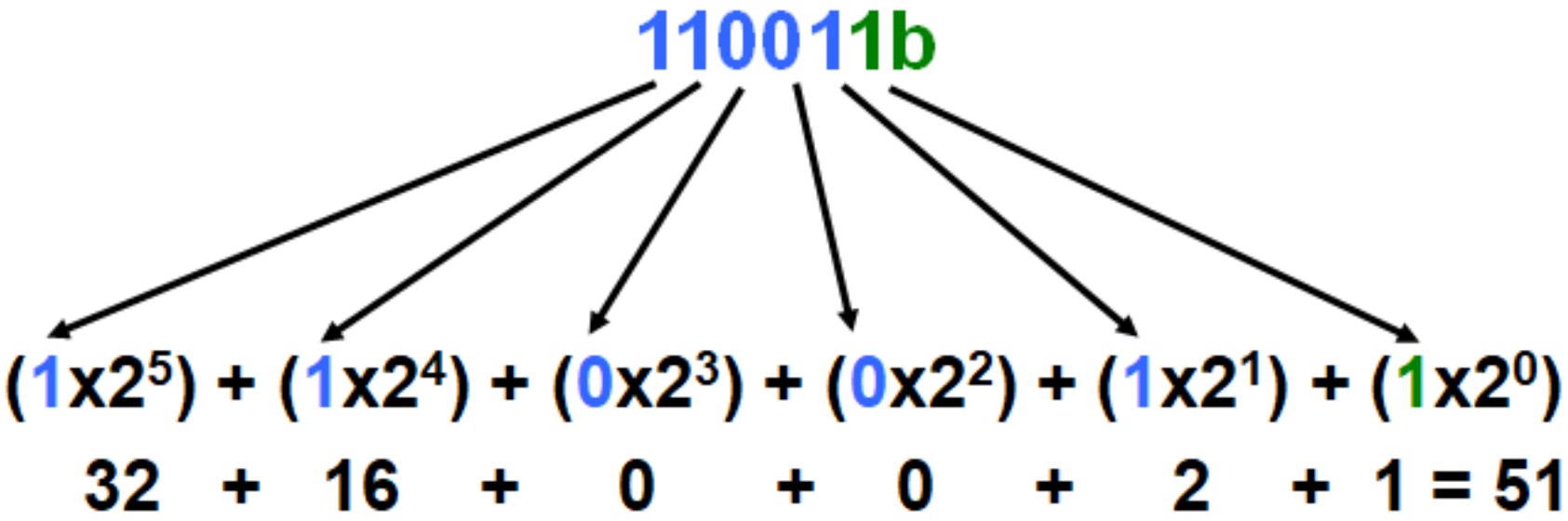
$$1011b = (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$$

Low order bit

High order bit

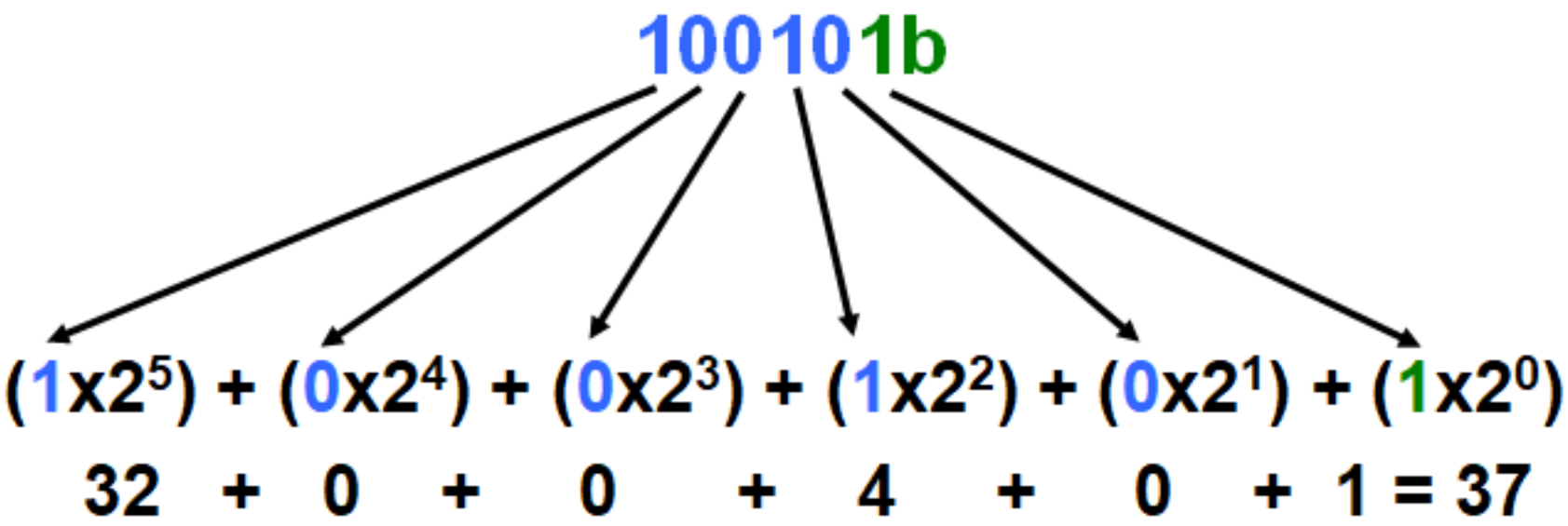
Binary Number Example 1

- What is the decimal (base 10) number represented by the binary number 0011 0011b?
- Solution:
  - Consider the power of 2 for each bit position
  - if there is a 1 bit then add that power of 2



Binary Number Example 2

- What is the binary number for the decimal number 37?
- Solution:
  - Consider the power of 2 for each bit position
  - if there is a 1 bit then add that power of 2



Decimal to Binary Number Algorithm

- What is the binary representation for the decimal (base 10) number 83? Express the answer in nibbles.

$$
\begin{array}{rcrl}
83/2 & = & 41 \text{ remainder } 1 & 1x2^0 \\
41/2 & = & 20 \text{ remainder } 1 & 1x2^1 \\
20/2 & = & 10 \text{ remainder } 0 & 0x2^2 \\
10/2 & = & 5 \text{ remainder } 0 & 0x2^3 \\
5/2 & = & 2 \text{ remainder } 1 & 1x2^4 \\
2/2 & = & 1 \text{ remainder } 0 & 0x2^5 \\
1/2 & = & 0 \text{ remainder } 1 & 1x2^6
\end{array}
$$

- most significant bit is the last one in the list.
- the binary number is 0101 0011 b

Lab Exercises

Express you answers in nibbles - 8 or 12 bits in groups of 4 with spaces between the nibbles.

Q2 How do you write the decimal value 19 in binary?

Q3 How do you write the decimal value 38 in binary?

Q4 How do you write the decimal value 47 in binary?

Q5 How do you write the decimal value 268 in binary?

Q6 Express the decimal integer 89 in an 8-bit unsigned binary number.

Q7 Express the decimal integer 298 in an 8-bit unsigned binary number.

Q8 What is the decimal value of the unsigned binary number 0010 0100 b?

Lab Exercises

Q9 What is the unsigned decimal integer value represented by the binary number 1010 1010 b?

Q10 What is the unsigned decimal integer value represented by the binary number 0001 0101 0100

Q11 What is the relationship between these two base 10 values?

Q12 Given your previous answer how do you multiply a binary number by 2?

Q13 How can you divide a binary number by 2?

Q14 What number is represented by the following binary numbers? 1, 11, 111, 1111?

Q15 What is the pattern of the numbers above?

Q16 What number is represented by 11 1111 1111 b?

How many ways are there to make a pizza?

- If there are 7 different toppings and every pizza includes cheese.
- For each pizza one bit represents one topping. A plain cheese pizza would be 0000000. The "works" would be 1111111.
- $2^7 = 128$ different combinations of toppings.
- What is we have 8 different toppings? How many possible pizza combinations could we have?
- Generalize it to n toppings? 2^n combinations

Lab Exercises

Work this in the opposite direction.

Q17 Suppose we want to have 16 different combinations of pizza toppings excluding cheese. How many toppings should we have?

Q18 Suppose we want 50 combinations excluding cheese. How many toppings should we have?

Q19 Generalize these results. How many bits do we need to allow for n combinations? You are going to have to play with this mathematically to find the right answer. A scientific calculator may help.

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Hexadecimal Notation

Long binary values are frequently expressed as Hexadecimal numbers.

Decimal	Binary
0	0000 b
1	0001 b
2	0010 b
3	0011 b
4	0100 b
5	0101 b
6	0110 b
7	0111 b

Decimal	Binary
8	1000 b
9	1001 b
10	1010 b
11	1011 b
12	1100 b
13	1101 b
14	1110 b
15	1111 b

Four binary digits = 1 hexadecimal digit.

Hexadecimal Notation

Each group of 4 bits gives us one hexadecimal digit.

Decimal	Binary	Hex
0	0000 b	0x0
1	0001 b	0x1
2	0010 b	0x2
3	0011 b	0x3
4	0100 b	0x4
5	0101 b	0x5
6	0110 b	0x6
7	0111 b	0x7

Decimal	Binary	Hex
8	1000 b	0x8
9	1001 b	0x9
10	1010 b	0xA
11	1011 b	0xB
12	1100 b	0xC
13	1101 b	0xD
14	1110 b	0xE
15	1111 b	0xF

By convention, 0x precedes hexadecimal digits.

Lab Exercises

Q20 How do you write 1100 1101 b in hexadecimal notation?

Q21 How do you write 1001 0011 1010 b in hexadecimal notation?

Q22 How do you write 0011 1110 b in hexadecimal notation?

Q23 How do you write 0xFAC7 in binary?

Q24 How many hexadecimal digits are needed to represent a 32-bit binary number?

Lab Exercises

Q25 Convert 1010 1011 1110 b to hexadecimal

Q26 Convert 1010 1100 1110 b to hexadecimal

Q27 Convert 1011 1110 1101 b to hexadecimal

Q28 Convert 1011 1110 1110 1111 to hexadecimal

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Example Decimal to Hex

- What is the hexadecimal representation for 1348 base 10?
- We can use the same repeated division strategy used to convert from decimal to binary algorithmically.

$$\begin{array}{rcrl} 1348 \ / \ 16 & = & 84 & \text{with a remainder of 4} \\ 84 \ / \ 16 & = & 5 & \text{with a remainder of 4} \\ 5 \ / \ 16 & = & 0 & \text{with a remainder of 5} \end{array}$$

- The last remainder is the high order digit just like the previous algorithm.
- The hexadecimal representation is 0x544.
- Check: it $(5x16^2) + (4x16^1) + (4x16^0) = 1280 + 64 + 4 = 1348_{10}$

Lab Exercises

Q29 What is the hexadecimal representation for 1082 base 10?

Q30 Is 0xF4C8E an even or odd decimal number?

Q31 Is 0x78D03B an even or odd decimal number?

Q32 How many bits are in the binary value for the hexadecimal number 0x87CE?

Hand in Your Lab

Save the document and submit it on Brightspace.